



Society of Petroleum Engineers

SPE-229888-MS

Redevelopment Strategy of a Reservoir Under Miscible Injection: Balancing Reservoir Performance and Resource Utilization

M. Y. Alklihi, N. Aljneibi, R. Alwaeel, S. Ajmal Haider, K. Khan, and A. Al Tenaiji, ADNOC Onshore

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229888-MS](https://doi.org/10.2118/229888-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

An integrated subsurface re-development strategy is presented in this work for a reservoir which has been under 20 years of miscible HC WAG injection and characterized by low permeability and large transition zone. One of the keys to developing such reservoir is to continuously ensure the establishment of an effective displacement system. The study aims to enhance the field's production and recovery through a series of phased projects, including conversion to full water injection, with a focus on optimizing well spacing. The redevelopment plan is anticipated to maximize oil recovery while minimizing dependency on HC gas consumption and maintaining reservoir pressure. The currently implemented development plan has been facing several complexities and challenges including maintaining miscibility conditions, sustainability of production and injection, associated operating costs and expansion of field development in reservoir extensions.

An assessment of multiple alternative field development options is conducted to tackle field management and reservoir challenges. The workflow is initiated by evaluation of historical reservoir management, HCWAG and CO₂ injection practices to identify areas for improvement and accelerate decision making process. The identified areas of improvement are incorporated into a dynamic model via diverse set of field management logics to screen a wide range of parameters. This involves running tens of simulation scenarios to evaluate converting the field to full water injection, CO₂-WAG, combinations of WI and CO₂-WAG or delayed HC/CO₂ WAG implementation.

The study also assessed the impact of well spacing (750, 375 and 250 m spacing) and higher offtake rates on production performance. Sensitivity analyses are performed to determine the optimal development setup, including vertical placement of wells and the use of maximum reservoir contact (MRC) wells. In the final step, optimal scenarios are compared to existing business plan and later selected, in line with strong economic indicators, honoring operational constraints, corporate strategic objectives.

The results are analyzed using standardized approach where several key indicators were cross-referenced to produce an optimal field redevelopment scenario with regards to balanced pressure depletion, harmonized sweep as well as robust discounted cash flow. The results indicate that delayed WAG scenarios can demonstrate comparable results to the WAG from day-1 scenarios, with a shorter production plateau but higher recovery. Full WI cases with a carefully studied ramp-up and shorter spacing can also produce an

economic scenario, as they offer a balance between production performance and resource utilization. De-risking activities are put in place to confirm the outcomes ahead of full field implementation.

The adopted workflow applied in this study uses a comprehensive dynamic modeling workflow to tackle reservoirs challenges relating to WAG performance, generating alternative development options and leveraging advanced field management logics.

Introduction

Integrated Reservoir Performance (IRP) analysis and Redevelopment Assessment is based on a combination of "Bottom Up" and "Top Down" approaches, with a focus on elements that have the greatest impact on the Field Development Plan (FDP) and Business Plan (BP). This method was previously described in detail by Tahir et al. in 2019.

The study described in this paper is divided into three comprehensive sections: Challenges, Uncertainties & Risk Mitigation, and FDP Optimization through Redevelopment Strategy from subsurface and surface aspects.

In the next section, we delve into the model status and robustness. This is to showcase the insights generated during various analyses performed throughout the study. We also examine the risks associated with the model's predictive capabilities, reviewing perspectives from multiple domains including geology, reservoir, and production. This holistic approach ensures that all relevant factors are considered, providing a robust foundation for the subsequent sections.

The paper also addresses the challenges faced in reservoir management. Here, we investigate practices such as Hydrocarbon Water Alternating Gas (HCWAG) and CO₂ injection. By thoroughly analyzing the uncertainties, we can develop strategies to mitigate risks and enhance the reliability of our predictions.

The study concludes with several optimized FDP scenarios, trying to see which redevelopment scenario can yield similar recovery factor to the existing plan, despite switching to water injection. This final product provides a range of options for decision-makers, allowing them to choose the most suitable strategy based on their specific goals and constraints.

Field Overview

The major reservoir of this study has been described in earlier work published in SPE-208044-MS. The reservoir is characterized by a faulted, low-relief, four-way anticline closure that contains oil. Two distinct patterns of strike-slip fault systems have been identified: the N75W and N45W conjugate fault systems. These systems are indicative of a strike-slip fault regime, as evidenced by the low ratio between fault lengths and throw, nearly vertical fault planes, and alternating structural highs and lows at fault relays within the fault zones (Salahuddin et al., 2017). These faults are considered open and do not create any compartmentalization within the reservoir.

Deposited during the Lower Cretaceous period, the reservoir lies within a carbonate platform environment and is overlain by a dense limestone interval, which serves as the cap rock. The deposition occurred over a vast carbonate ramp and can be divided into three systems tracts: the transgressive system tract, where wackestone and packstone with some orbitolinas and algal mounds were deposited; the early high system tract, where wackestone to rudstone were deposited; and the late high stand systems tract, where grainstone and rudstone with some miliolids and rudists were deposited. There is good coverage of core data, providing sufficient information on thin section analysis and reservoir heterogeneity.

The reservoir is further subdivided into different subzones separated by sub-dense units. These sub-dense layers represent minor flooding events within the porous units, indicating shallowing phases between the porous subzones, which exhibit variable porosity and permeability ranges. Generally, there is very low permeability contrast across the different subzones, and the reservoir is considered to have relatively low permeability.

The reservoir exhibits better characteristics at the crestal area, with properties gradually deteriorating towards the flanks. It consists of tight to moderate carbonates with permeability ranging from 5 to 10 mD. The oil within the reservoir is light.

The reservoir was put on an Early Production Scheme (EPS) in 1994 for data gathering and evaluation, following successful appraisal outcomes. Phase-I Development was defined and implemented in 2006, incorporating Miscible Water Alternating Gas (WAG) injection with rich gas from day one, based on careful screening and experimental results. Subsequent development phases followed, and as of today, the entire reservoir is producing under miscible WAG.

The reservoir has been producing for almost 20 years under miscible HC-WAG injection. It was decided to implement HC-WAG EOR since day-1 of production with the aim of enhancing oil recovery and increasing the oil production in a sustainable way. Based on the reservoir properties, some of the most important aspects of the injection project are:

- Rich gas is injected at high pressures in a multi-contact miscible flood mechanism.
- Gas and water slugs are injected in a 6 by 6 months cycle, where the gas and water rates are adjusted to meet a voidage replacement ratio of 1.0.
- Initial phase of development followed a five-spot pattern while the subsequent phases followed line-drive pattern.
- Production is constrained by a minimum bottomhole pressure which does not fall below saturation pressures to avoid gas liberation.

Overall, water cut and GOR gradual development are observed from the field, while the simultaneous voidage replacement ratio (VRR) is maintained around 1.0-1.2. The figure below represents the historical production performance of the reservoir.

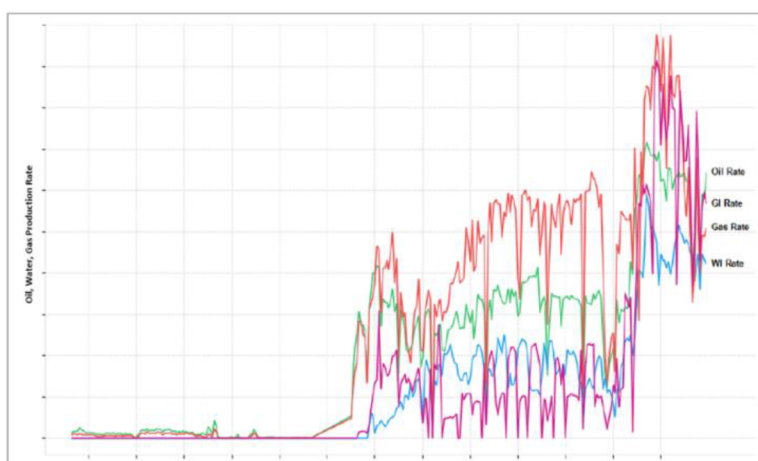


Figure 1—Historical production performance of the field reflecting three (3) phases of development all under miscible WAG injection (x-axis is time in years since commissioning).

Objectives

The reservoir's future development plan considers several EOR Techniques to be implemented in flank areas as part of production growth plans including CO₂ WAG. In the core area, infill drilling and partial conversion from HC WAG to CO₂ WAG is planned. Following corporate strategy to minimize dependency on HC gas consumption, several redevelopment options were evaluated involving different infilling options, full WI and delayed WAG.

In another words, the redevelopment evaluation aimed to minimize gas consumption while maximizing oil production. We evaluated strategies such as replacing HC-WAG/CO₂ with water injection and reducing

well spacing to less than 400 meters or even 250 meters and utilization of Maximum Reservoir Contact wells (MRC) to sweep bypassed oil. A proposal for production growth implementation with delayed WAG for 5-7 years is generated, implying full field conversion to WI and later switch back to WAG. This strategy, which showed excellent performance in terms of reservoir target delivery and recovery factor, allows maximization of HC Gas can be exported for domestic purposes while reducing CAPEX and OPEX.

Redevelopment Strategy

To achieve the targets set out in the production growth strategy along with overcoming current challenges of the existing development scheme, it is essential to support business plans with practical continuous improvement approaches. These approaches should aim to explore alternatives that can deliver optimal development plans and unlock additional value.

The current Field Development Plans (FDPs) aimed to enhance production and align with the corporate's strategy. The optimal development plan focused on three main strategies: water injection, rich gas as water-alternating-gas (WAG) injection with six-month gas cycles followed by six months of water, and the option to switch to three-month gas and six-month water cycles or CO₂ WAG in the future. These strategies were successfully implemented.

In this study, we highlight our adopted strategies for production growth while minimizing costs. Our redevelopment evaluation aimed to minimize gas consumption while maximizing oil production. We evaluated strategies such as replacing HC-WAG/CO₂ with water injection and reducing the well spacing by infill drilling in the core development area and reduced spacing in the future development area.

The study involves screening of different complex processes such as WAG, CO₂, CO₂ WAG injections, therefore it is important to have proper grid resolution to minimize numerical dispersion effects, increase accuracy of flow dynamics and maintain practical simulation runtimes. Therefore, in order not to compromise on physics and simulation runtimes, the high-resolution reservoir simulator was used with a high-resolution grid of 50×50m cell size in the development areas.

Pattern Drilling with Different Well Spacing

Infill drilling remains one of the most traditional and effective methods to enhance production in oil reservoirs. To meet the increased production targets, new areas of the reservoir, particularly the North, South, and East flanks, were explored for further development through infill drilling. This approach aims to maximize the extraction of remaining recoverable oil by strategically placing new wells in areas with high potential. Building on earlier development strategies, an automated approach was employed to screen and identify areas with significant remaining recoverable oil potential. This automated screening process involved evaluating incremental recovery through line-drive well designs, which alternate between one Water-Alternating-Gas (WAG) injection well and one production well. This method ensures a balanced and efficient extraction process, optimizing the recovery of hydrocarbons.

To further refine the development plan, sensitivity analyses were conducted to optimize the performance of the well patterns by reduced spacing. These analyses considered various factors, such as reservoir properties, fluid dynamics, and operational constraints, to ensure the optimal placement and operation of the wells. The identification of high-potential infill drilling spots was achieved by generating a map of a modified opportunity index (OI), as described by Mustapha & Dias in 2018. This index takes into account various geological and reservoir parameters to highlight areas with the greatest potential for incremental recovery. By using this comprehensive and data-driven approach, the development team was able to pinpoint the most promising locations for infill drilling, thereby enhancing the overall efficiency and effectiveness of the reservoir development plan.

$$OI = 0.31548 \cdot NTG \cdot DZ \cdot (S_{oil} - S_{or}) \cdot \sqrt[2]{k \cdot \phi} \cdot \sqrt[3]{\frac{P_{init}}{P_{init} - P_{current}}}$$

The OI map was used to highlight areas with high potential for further development by pattern infill drilling. The well placement and selection process consisted of the following key steps, and examples are shown in the figures below:

- Identify high potential recovery areas.
- Generate line-drive patterns of wells alternating injection (I) and production (P)
- Ensure maximum coverage of the area, depending on well length and spacing.
- Perform sensitivity based on group production potential.
- Identify uneconomic wells after each sensitivity and discard them.

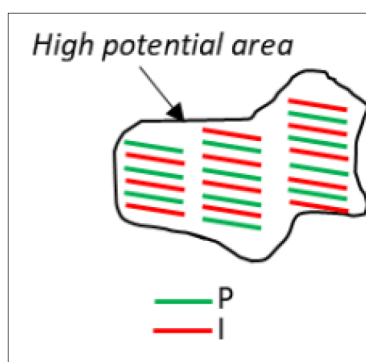


Figure 2—Example of a high potential area with alternating injection and production wells

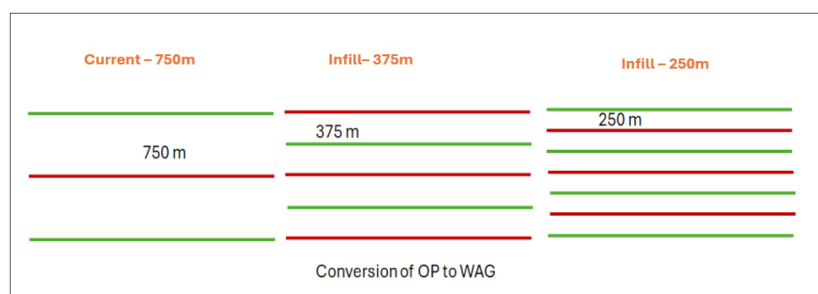


Figure 3—Schematic of well patterns with reduced well spacing.

The potential production from new wells is also sensitized. We used a set of smart workflows to automate well placement, completions, and generation of range of simulation models for screening purposes. The sensitivities revealed that optimum spacing between wells in a pattern is 375m. This layout, for various well length, is therefore selected for the well placement strategy optimization.

A full water injection case was developed considering reduced spacing to 375 m and another case with further reduction to 250 m spacing.

The impact on new well count was increased by more than 1.7 times in the case of 250 m spacing while the added value in terms of recovery factor was less than 1.5% relative to the 375 m spacing case. In efforts to try to improve the results further, a ramp-up was investigated considering both reduced spacing but the results always showed that further reduction to 250 m spacing is not very attractive. This is attributed to excessive water production and breakthrough. Below figures show the impact on sweep efficiency by reduced spacing to 375 m spacing.

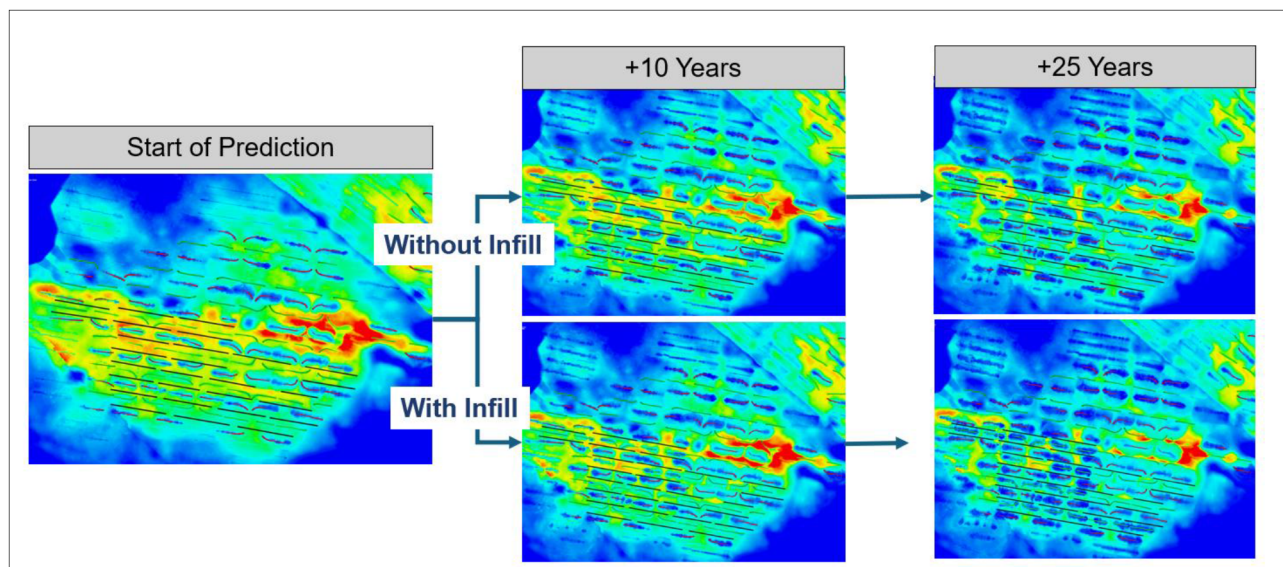


Figure 4—Impact of reduced spacing on sweep efficiency.

Injection scheme

In order to reach a conclusive outcome on the redevelopment injection scheme, which walks away from HC injection, a structured approach was followed:

Table 1—Sensitivity on injection scheme

Case	Description
Reference	Existing development
Case-1	Full WI: Convert the entire field to WI
Case-2	Full CO ₂ -WAG: Switch HC-WAG wells to CO ₂ -WAG wells
Case-3	Partial WI / CO ₂
Case-4	Full WI followed by WAG: Switch the field to WI and later come back to EOR scheme

Along with the injection scheme, several other parameters were also investigated to maximize oil recovery including, but not limited to:

- Production ramp-up by acceleration of infill drilling.
- Vertical well placement sensitivity to investigate the effectiveness of the currently adopted completion strategy.
- Gas lift sensitivity to assess artificial lift requirements for the water injection cases.
- Delayed WAG concepts by switching the field entirely to water injection for few years and then return back to WAG/EOR to ensure minimum impacts on ultimate recovery.
- Gradual decrease in gas injection requirements with time.

For this paper, we will present part of the sensitivity and focus on high level outcomes.

The figure below shows the differences between the 3 different injection schemes. Compared to WAG cases, water injection showed 3-4 years shorter plateau corresponding to 2-3% lower RF.

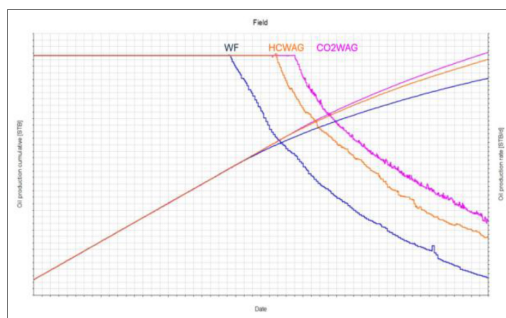


Figure 5—Comparison of water injection vs. rich HC-WAG injection vs. CO2 injection vs. Time (Years) considering same number of wells.

Ramp-up feasibility from full water injection scenario

Ramp-up options were investigated in efforts to improve the water injection scenario and bring it as close as possible to the gas injection existing development scenario. The outcomes of this scenario were not very attractive such that even with ramp-up, the recovery factor was always less by 2-3% compared to HC-WAG.

Delayed WAG case

In this scenario, it was assumed that the full field will be switched to water injection in order to maximize the gas export for a certain period of time (3 to 10 years of water injection) and then switch back to WAG in a way that the least impact is attained on ultimate recovery factor. The results of this scenario were encouraging. Refer to the figure below.

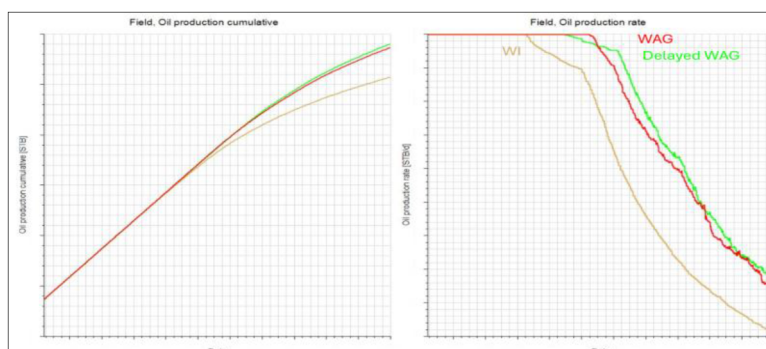


Figure 6—Comparison of WAG vs. Delayed WAG concept vs. Time (Years).

In principle, it was investigated to implement WAG within 2 years head of the plateau decline. Additional sensitivities were made to the delayed WAG scenarios by accelerating some infill to ramp-up additional production and considering the different schemes. Refer to the figure below.

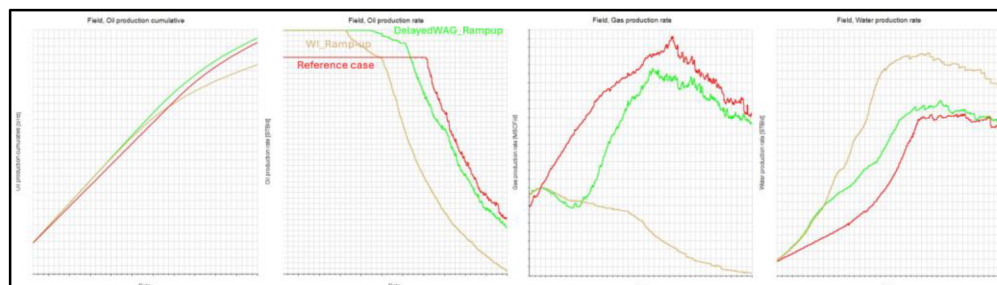


Figure 7—Comparison of WI vs. Delayed WAG concept considering Ramp-up vs. Time (Years).

Gas lift requirements due to redevelopment by water injection

Over time, water breakthrough or insufficient pressure support can impair and limit the productivity of certain wells, potentially leading to a decline in production due to ongoing injection operations. Gas lift injection can enhance well productivity by reducing the fluid mixture's density in tubing. The costs associated with gas lift (GL) injection operations must always be justified by the additional oil produced.

Several scenarios are compared varying GL constraints per well:

- Scenario-1: Zero gas lift assigned for all new wells.
- Scenario-2: Fixed gas lift injection rate per well with varying quantities.
- Scenario-3: Optimized gas lift through gas lift optimizer (GLO) algorithm i.e. only injection of gas lift when the field cannot meet its target.

The outcomes of the above scenarios clearly demonstrate the necessity of gas lift implementation in all the cases (refer to the figure below). In the final evaluated development scenarios, scenario-3 was followed which relies on an algorithm to optimize the allocation of gas lift (GLO) to the wells that benefit the most, by assessing the minimal economic gradient, thereby minimizing costs associated with excessive gas lift injection (Ismail A. M. M., et. al., 2020).

The GLO algorithm determines the optimal amount of lift gas to allocate to each well to achieve the field production targets. The optimization process involves the following steps:

- Divide the total lift gas supply into uniform amounts.
- Add lift gas increments to the wells that show the most significant oil gain with increased lift gas.
- Remove lift gas from the wells that exhibit the least benefit, i.e., those that lose the least oil with decreased lift gas.

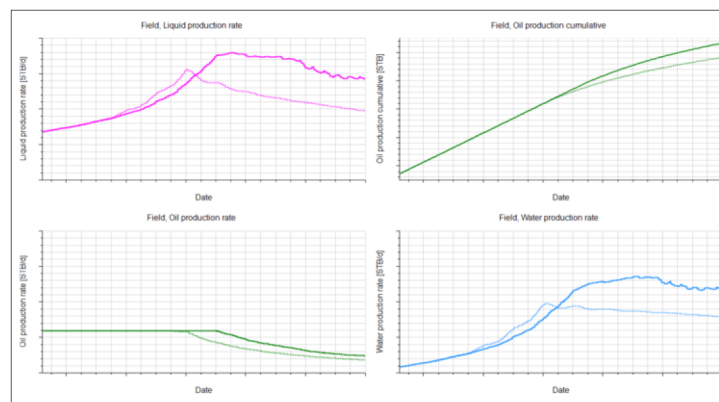


Figure 8—Comparison of zero gas lift scenario (dotted) vs. optimized gas lift scenario (solid) vs. Time (Years).

Surface Facilities Assessment

Case-1: Full Water Injection (WI)

Several facilities are required. These include water source either from Aquifer, sea or produced water treatment facilities, which involves coarse and fine filtration, de-oiling hydro cyclones, and de-aeration towers or membrane contactors. High-pressure water injection pumps with variable frequency drives (VFDs) are necessary, along with booster pumps at headers and corrosion-inhibited lined flowlines. New WI manifolds, pressure control valves, and zonal injection metering are also needed. Additionally, produced water separation, reinjection or disposal units, surge tanks, and sand management systems are essential.

In terms of design and safety considerations, oxygen scavenger dosing, a scale and corrosion inhibitor program, and continuous monitoring of dissolved oxygen (DO), iron, and salinity are crucial. Redundancy in injection pumping and surge protection is required to mitigate water hammer. Corrosion-resistant metallurgy and pressure relief systems should be incorporated into the WI network.

Operational issues include managing high watercut and the increased load of produced water handling. Ensuring the reliability of injection pumps and addressing the high energy demand are also important. To mitigate these issues, preventive maintenance and spares management for pumps are necessary. Optimizing produced water recycling and implementing continuous chemical treatment and online monitoring can help reduce scaling and corrosion risks.

Case-2: Full CO₂-WAG: Switch HC-WAG wells to CO₂-WAG wells

To implement the necessary facilities for CO₂ capture and import, several components are required. This includes dehydration systems such as molecular sieves, glycol, or membranes to remove the saturated water. Multi-stage CO₂ compressors with intercooling and dense-phase pumps are essential, along with a corrosion-resistant CO₂ distribution network and injection manifolds. Additionally, water injection facilities are necessary. The design and safety considerations involve using corrosion-resistant alloys (CRA) or coated metallurgy, CO₂ corrosion monitoring, and leak detection systems. Specialized health, safety, and environmental (HSE) requirements include ventilation, gas detection, and hazardous area classification. Custody transfer metering and online analyzers for CO₂ purity and H₂O content are also crucial. Operational issues to consider include corrosion and equipment integrity risks, CO₂ breakthrough, and recycle capacity limitations, as well as compressor downtime and availability constraints. To mitigate these issues, continuous CO₂ corrosion inhibitor injection and frequent inspection campaigns are necessary. Compressor redundancy and preventive maintenance programs should be implemented, along with real-time monitoring for CO₂ leakage and breakthrough control.

Case-3: Partial WI / CO₂ Hybrid

Hybrid model allows selective water injection (WI) and CO₂ injection at the pad level. This requires a medium-capacity WI package that includes filtration, de-aeration, and high-pressure pumps. Modular CO₂ compression and dehydration skids with a scalable design are also necessary. Additionally, flexible produced fluid handling systems are required, which include additional three-phase separation, CO₂ recycling, and produced water reinjection. The design and safety considerations involve a flexible control philosophy that allows for the safe alternation between WI and CO₂ injection. Allocation metering is needed to monitor the injection split, and dual chemical programs must be tailored for alternating WI and CO₂ service. Operational issues include the complexity of switching between WI and CO₂ operations, fluctuating corrosion regimes, and potential operational errors. To mitigate these issues, automation and interlocks for switching modes are necessary, along with operator training on hybrid operations. Continuous corrosion monitoring and adaptive chemical dosing are also essential to ensure the system's integrity and efficiency.

Case-4: WI followed by CO₂ WAG

To embark on the journey of water injection (WI) followed by water alternating gas (WAG) injection, a significant initial investment is required. This includes a complete WI train encompassing water treatment, high-pressure pumps, and distribution systems. Provisions for future CO₂ facilities, such as compressors, dehydration units, and manifolds, must be integrated from the outset. The produced water handling capacity must be designed to accommodate the WI plateau, retrofitting CO₂ recycle facilities and automation for WAG cycles is essential. The design must be future-proof, with spare plot space, foundations, and electrical provisions in place. Critical tie-in points should be upgraded with corrosion-resistant alloys to handle CO₂ service later on. The safety design must be dual phase to manage both WI now and CO₂ in the future. Operational challenges include the concentration of capital expenditure (CAPEX) during the WI phase and

the risks and downtime associated with retrofitting. To mitigate these issues, a staged CAPEX strategy can defer CO₂ investment, and pre-designed tie-in points can minimize downtime during retrofitting. Early operator training and transition planning for the WAG phase are crucial to ensure a smooth and efficient operation.

Conclusion

The study presents an integrated redevelopment strategy for a low-permeability reservoir under 20 years of miscible hydrocarbon water alternating gas (HC WAG) injection, aiming to optimize production and resource use by evaluating alternative injection schemes and well spacing. The reservoir has operated under miscible HC WAG for two decades, facing challenges in maintaining miscibility, sustainable production, injection, and operational costs. The redevelopment aims to enhance recovery while reducing hydrocarbon gas use and sustaining reservoir pressure.

- **Dynamic modeling and scenario screening:** Multiple development options were assessed through dynamic reservoir simulation, including full water injection (WI), CO₂ WAG, and combinations thereof, alongside sensitivity analyses on well spacing and production rates to identify optimal redevelopment scenarios.
- **Well spacing and infill drilling:** Sensitivity analyses showed that reducing well spacing to 375 m optimizes recovery and production efficiency, while further reduction to 250 m increases well count significantly but yields limited incremental recovery due to water breakthrough and excessive water production.
- **Injection scheme comparisons:** Full water injection scenarios show a 3–4 year shorter production plateau and 2–3% lower recovery factor than HC WAG, while delayed WAG (initial water injection followed by WAG) can achieve comparable ultimate recovery with operational and economic benefits.
- **Gas lift optimization:** To counteract production declines from water breakthrough or pressure loss, a smart gas lift optimization algorithm allocates lift gas economically to wells, enhancing productivity while minimizing operational costs.
- **Surface facilities:** each of the four cases presents unique requirements and challenges for implementing water injection (WI) and CO₂ injection systems. Case-1 focuses on the comprehensive setup for full WI, emphasizing the need for extensive facilities, design and safety considerations, and operational management. Case-2 highlights the transition to full CO₂-WAG, requiring specialized equipment and addressing corrosion and integrity risks. Case-3 offers a hybrid approach, allowing selective WI and CO₂ injection, which demands flexibility in design and operations. Finally, Case-4 outlines a phased strategy for WI followed by CO₂-WAG, necessitating significant initial investment and future-proofing the design to accommodate both phases. Each approach requires careful planning, robust design, and proactive operational management to ensure efficiency and safety.

Acknowledgement

The authors would like to thank the field operator for their support and permission to publish this paper. In addition, the authors would like to acknowledge other members of the team and its affiliates who have contributed to the development of this study.

Nomenclature

CO₂ Carbon Dioxide
DZ Cell Height

GL	Gas Lift
GLO	Gas Lift Optimizer
HC	Hydrocarbon Gas
K	Permeability
mD	Millidarcies
NTG	Net to Gross Ratio
OI	Opportunity Index
Φ	Porosity
P_{init}	Initial Reservoir Pressure
$P_{current}$	Current Reservoir Pressure
S_{oil}	Oil Saturation (Current)
S_{or}	Residual Oil Saturation
WAG	Water Alternating Gas
WI	Water Injection

References

- Alklih, M. Y., Salahuddin, A.A., Khan, K. A., Aljneibi, N. M., Rat, C., Ramatullayev, S., Su, S., Alfonzo, A., Mustapha, H., Mihai, M., (2021) "Integrated Subsurface Study For Gas-Oil Communicating Reservoirs: From Structural Synthesis To Optimum Co-FDP", SPE-208044-MS, Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE.
- Ismail A. M. M., Alnaqbi J. M. R., Mustapha H., Su S., Alfonzo A., (2020) "Efficient and Flexible Reservoir Field Management Gas Lift Optimization Approach to Unlock the Production Potential and Economic Investment of Giant Carbonate Reservoir", SPE-203092, Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE
- Salahuddin, A.B., Latief, A. I, Zaghum, U.Q., Al Shehhi, A. S., Al Hammadi, K. E., (2017) "Static and Dynamic Uncertainty Management for Probabilistic Volumetric and Production Forecast: A Case Study from Onshore Abu Dhabi", SPE-186078, SPE Reservoir Characterisation and Simulation Conference and Exhibition, Abu Dhabi, UAE